Coherent Possibility and Selective Actualization: A Two-Level Framework for Foundational Explanation

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Abstract

A structural or mathematical account can explain admissibility, but it does not yet explain why a determinate world is actual, why actuality is historically articulated, or how becoming proceeds without appeal to a brute selector. This paper argues that the resulting gap is best understood as a problem for foundational explanation. Any adequate foundational account of a nontrivial, dynamically stable, historically articulated world must distinguish two irreducible explanatory roles. The first is constitutive constraint: the conditions under which a candidate structure counts as coherently realizable at all. The second is selective actualization: the conditions under which one admissible continuation rather than another becomes integrated into realized history. I develop the constitutive role in presentation-independent terms through coherent realizability, equivalence-invariance, canonicity, and local satisfiability of coherence burdens. I then propose a localist account of the selective role in terms of commitment, admissibility, local generative affordance, stabilization burden, sufficiency, primeness, and integration. The aim is not to derive a complete ontology or a finished dynamics, but to clarify the explanatory architecture that any adequate foundational explanation must exhibit and to distinguish argued constraints from stronger proposals and deferred formal development.

Keywords: actualization; laws of nature; structural realism; modality; process ontology; philosophy of science

1 Introduction: the problem of actualization

Accounts of reality that emphasize structure, mathematics, or modal order promise an explanation of what kinds of world are admissible. They do not yet explain why one admissible continuation is actual, why actuality is historically articulated rather than merely catalogued, or how determinate becoming can proceed without appeal to a brute selector. The gap between admissibility and actuality is the problem addressed in this paper.

The attraction of structural or mathematical metaphysics is nevertheless real. If reality has any intelligible order at all, then it must answer in some way to structure. The hope behind the Mathematical Universe Hypothesis is that the world is not merely *described* by mathematics, but *is* mathematics in a sufficiently robust sense.[1] That family of views captures an important point: whatever exists must at least be compatible with the order of coherent form. The difficulty is that form alone seems to explain, at most, a space of coherent possibility. It does not by itself explain

why one possibility is actual, why actuality unfolds through determinate transitions, or why realized history is more than an undifferentiated collection of admissible forms.

The central claim advanced here is therefore not that a final ontology has already been reduced to a definitive pair of laws. It is instead that any adequate foundational explanation of a nontrivial, dynamically stable, historically articulated world must distinguish two irreducible explanatory roles. The first is *constitutive constraint*: it determines what can count as a coherently realizable candidate for existence at all. The second is *selective actualization*: it determines how one admissible continuation rather than another becomes integrated into realized history. The paper argues that these roles are explanatorily distinct even when any later formal theory may articulate them by many principles rather than two single axioms.

The resulting framework can be stated compactly:

> **Constitutive constraint:** the explanatory role that fixes admissible structure.
> **Selective actualization:** the explanatory role that governs actual integration into history.

The second role is *local* in a specific sense that will matter throughout. Locality here is not spatial locality. It means locality to the present continuation cone of realized history: actualization must proceed by features readable in the realized present and its immediately admissible continuations, rather than by a prior survey of indefinitely remote futures. The proposal is thus path-dependent without becoming globally foresighted.

This paper is a contribution to foundational explanation in philosophy of science, not a complete metaphysical system. It does not introduce a finished evaluator, a cosmology, or the full PEN machinery. Its aim is more limited and, for present purposes, more useful: to identify the explanatory architecture required if a determinate world is to be both admissible and actualized, to formulate restrained candidate statements of the two roles, and to distinguish argued constraints from stronger proposals and deferred formal development.

The immediate target is not every conceivable universe whatsoever, but any world stable enough to preserve determinate identity through change, dynamic enough to exhibit genuine becoming, and rich enough to sustain nontrivial historical articulation. On that target, the paper proceeds as follows. Section 2 situates the proposal among familiar distinctions and identifies the remaining explanatory gap. Section 3 argues that foundational explanation must separate constitutive admissibility from selective actualization. The subsequent sections develop each role in turn, clarify what the framework adds, and conclude by distinguishing what is argued, what is proposed, and what is deferred.

2 Existing distinctions and the remaining gap

Several familiar distinctions already mark genuine structure in debates about laws, modality, and world-description. One can distinguish a space of states from a dynamics defined on that space, laws from initial conditions, structural description from realized history, and local evolution from whole-history or global-consistency formulations. These contrasts are indispensable, but they do not yet resolve the present problem. Even where they explain how a world may be represented, evolved,

or globally characterized, they do not by themselves explain why an admissible world is actual or how actualization is articulated from within realized history.

This is why the distinction proposed here is not simply a relabeling of state space and dynamics or of law and initial conditions. Constitutive constraint concerns what can count as a coherently realizable candidate for existence in the first place. Selective actualization concerns the integration of one admissible continuation rather than another into realized history. Those questions cut across more familiar distinctions because they arise at the level of foundational explanation rather than at the level of already-given physical models.

The same point applies to structuralist and holistic pictures. Structural realism and allied mathematical views illuminate why invariant form matters, but structural description alone does not explain actuality. Whole-history or global-consistency views illuminate the idea that local becoming may answer to wider constraints, but they do not remove the need to explain how actuality can be locally articulated without appeal to an external survey of indefinitely remote futures. The gap that remains is the one this paper targets: the gap between admissible structure and actualized history.

The present argument therefore concerns foundational explanation rather than any privileged implementation. It is not the claim that reality is literally "written" in any one human formalism, still less that the world is a proof assistant or that its source code can be identified with a particular syntactic calculus. Human logical systems are at best presentations of deeper invariant content. They matter here only insofar as they provide partial insight into what constitutive admissibility and selective actualization would have to preserve. Constructive type theory is useful because it treats existence as witness-bearing and because it makes the passage from possibility to actuality formally visible in the judgmental form $\Gamma \vdash a : A$.[2] Homotopy type theory is useful because it exhibits a presentation in which identity carries internal structure and equivalence can be elevated to ontological significance.[3] But the paper does not assume that reality is identical with either presentation. It seeks the invariant content beneath them.

Talk of a foundational law in what follows is shorthand for a role in foundational explanation, not yet a contingent regularity of a particular physical cosmos. The paper does not introduce a specific evaluator, local-actualization rule, or cosmological mechanism. It argues only that foundational explanation must separate two roles and that these roles must have a certain structural shape. It also does not treat arithmetic as a third primitive; stagewise language is used only as a weak convenience for speaking about cumulative history.

3 A Two-Level Framework for Foundational Explanation

3.1 The insufficiency of constitutive order alone

Suppose there were only a constitutive order: a realm of coherent forms, structures, or possibilities. Then one of three things must happen. Either nothing becomes actual, in which case no world exists. Or every coherent possibility is equally actual, in which case actuality is indistinguishable from undifferentiated possibility and there is no determinate history. Or one possibility is actual rather than another for no reason internal to the ontology, in which case actuality is brute.

None of these options is satisfactory. The first yields no world. The second yields no articulated

world. The third abandons explanation precisely where ontology should begin. A constitutive order alone is therefore incomplete.

3.2 The insufficiency of actualization alone

Now suppose there were only an order of actualization. An actualizing principle presupposes something to be actualized. If there is no prior distinction between coherent and incoherent, admissible and inadmissible, meaningful and meaningless, then the very idea of actualization is empty. An "actualization" that can equally well realize contradiction, incoherence, or indeterminacy is no law of reality at all.

Hence actualization cannot be primitive in isolation. It presupposes a prior order of coherent possibility.

These paired insufficiencies support the paper's core thesis. Any adequate foundational explanation of a determinate and historically articulated world must distinguish at least two irreducible explanatory roles: one that constrains what can coherently count as admissible structure, and one that governs selective actualization from among admissible continuations. The claim is architectural rather than axiomatic. It concerns the explanatory work that must be done, not the final number of principles in a completed formal theory.

3.3 Why putative third roles do not add a new explanatory level

One might object that a full ontology requires more than two primitive laws. Perhaps one law governs logic, another dynamics, another value, another temporality, and so on. At later levels of formalization that may be true. But at the foundational level the space of roles is exhausted more quickly.

At the level of foundational explanation, any putative further primitive role still appears to do one of two things. Either it constrains what can count as a coherent candidate for existence, or it governs the relation between such candidates and realized history. A principle of temporality, for example, functions constitutively insofar as it constrains the admissible form of temporal structure and selectively insofar as it governs actualization from within such structure. A principle of value functions selectively when it orders admissible continuations. A principle of identity functions constitutively when it fixes what counts as the same structure across variation.

The point, then, is not that a finished ontology will contain only two axioms. It is that the explanatory architecture of such an ontology must separate these two roles. Putative third roles are best understood as species of one or the other at the foundational level.

4 Constitutive Constraint

4.1 Why possibility must be coherent

The minimal requirement on a possible world is not merely that it be thinkable, but that it be *coherently realizable*. An entity or process cannot count as genuinely possible if its very being

depends on unresolved contradiction. A square circle does not fail because it is unusual; it fails because the conditions required to specify it cancel one another.

Yet bare non-contradiction is too weak. A world is not constituted by a list of propositions that happen not to clash. It is constituted by structures that must compose, persist, and interact. Thus possibility must be understood as *coherent realizability*: the mutual satisfiability of the structure's internal and external conditions of being.

Definition 1 (Coherent realizability)**.** *A candidate structure is coherently realizable when its constituents, relations, and operations can be jointly determined without unresolved contradiction or indefinite suspension of compatibility conditions.*

4.2 Identity, equivalence, and invariance

A second constitutive requirement concerns identity. If reality is structured, then sameness cannot be treated as a merely flat predicate detached from the internal relations that make a thing what it is. The identity of a structure must be rich enough to support the preservation and transport of its organization across admissible variation.

This does not yet force any one formal doctrine of identity, but it does force a principle of *equivalence-invariance*. If two presentations differ only in ways that make no invariant structural difference, then the difference cannot be ontologically fundamental. Reality cannot count as different what differs only by representational dressing.

Definition 2 (Equivalence-invariance)**.** *A constitutive order is equivalence-invariant when representationally distinct but structurally equivalent candidates are not counted as different in being.*

This principle is important because without it ontology becomes hostage to presentation. The same world would be multiply overcounted merely by redescribing it. A genuine constitutive law must cut deeper than syntax.

4.3 Canonicity and executability

A third requirement is canonicity. A possible world cannot be built from permanently stuck pseudo-objects whose closed realization never becomes determinate. If a construction is genuinely closed, reality must determine what it is. Otherwise the candidate remains an unresolved schematic placeholder rather than an executable world-constituent.

Definition 3 (Canonicity)**.** *A constitutive order satisfies canonicity when closed constructions admit determinate realization rather than remaining indefinitely non-canonical.*

The language of executability is not meant computationally in any narrow machine sense. The point is ontological: a world must be capable of running as a world. A purely schematic form that never resolves into determinate being is not enough.

4.4 Local satisfiability of coherence burdens

A fourth requirement concerns scale. Any genuine structure carries compatibility conditions. If these coherence burdens can grow without local discharge, then no stable world is possible. There would be only an ever-accumulating debt of unresolved compatibility, and reality would collapse into non-executability.

Hence a constitutive law must require that coherence burdens be *locally satisfiable*. New structure may introduce new conditions, but those conditions must be dischargeable within the world rather than indefinitely deferred.

Definition 4 (Local satisfiability)**.** *A constitutive order satisfies local satisfiability when the compatibility burdens introduced by new structure are locally dischargeable rather than requiring indefinitely unresolved global repair.*

4.5 Canonical formulation

We may now state the constitutive law in its strongest compact form.

Theorem 1 (Constitutive Law)**.** *A universe can exist only as a coherent, equivalence-invariant, canonically executable structure whose identity and compatibility conditions are locally satisfiable.*

This statement is intentionally presentation-independent. It does not identify the law with any one proof theory or any one formal semantics. Rather, it names the invariant content that any adequate presentation must preserve.

4.6 Why weaker alternatives fail

The constitutive law has the shape above because weaker candidates fail.

Bare consistency is too weak. One can imagine a body of non-contradictory descriptions that nevertheless never becomes an executable world because its constructions remain non-canonical or its compatibilities remain unresolved.

Flat identity is too weak. If sameness is treated as a purely extensional yes-or-no fact detached from structural transport, ontology becomes blind to the internal articulation of identity and loses the ability to distinguish genuine sameness from accidental coding.

Representation-sensitive ontology is too weak. If structurally equivalent presentations count as different worlds, being becomes hostage to syntax.

Unbounded coherence debt is too weak. If new structure may indefinitely accumulate unresolved compatibility burdens, then no stable world can be realized.

Taken together, these failures force the stronger constitutive statement above.

5 Selective Actualization

Constitutive constraint yields a realm of coherent possibility. It does not yet yield reality as realized history. The task of selective actualization is therefore not to replace the constitutive order, but to govern how actuality proceeds from within it.

Let H denote realized history at some stage. Let $\mathcal{C}_{\mathrm{loc}}(H)$ denote the local continuation cone of coherent continuations of H permitted by constitutive constraint. Selective actualization is a rule on $\mathcal{C}_{\mathrm{loc}}(H)$. The question is: what structural features must such a rule possess if it is to generate a nontrivial, dynamically stable, historically articulated world without hidden global foresight?

5.1 Why actuality must be local

A determinate universe cannot require an external oracle that surveys indefinitely remote futures before each act of becoming. If actuality depended on such a survey, the law of becoming would either be teleological in formulation or computationally opaque in implementation. Any viable Selective Law must therefore be *localizable*: it must rank candidate continuations by features available in the realized present and its immediately accessible continuation cone.

This locality requirement does not deny that present actualization has future consequences. It denies only that the law must precompute those consequences before actuality can proceed.

5.2 The Localizability Requirement

Definition 5 (Localizability Requirement)**.** *Any viable Selective Law must order admissible continuations by properties encoded in the continuation itself, in the realized history it inherits, and in the immediately opened field of further coherent continuations.*

Selective actualization may therefore be path-dependent without becoming globally foresighted. It reads what is already structured into the present continuation cone.

5.3 Commitment

Reality must preserve realized history. What has become actual cannot simply vanish from ontological account. This gives the Selective Law a path-dependent context without requiring global foresight: each continuation is judged relative to a cumulative inherited world.

Definition 6 (Commitment)**.** *A selective order exhibits commitment when realized history is cumulative: what has become actual remains part of the world's history and conditions future actuality.*

Commitment does not mean that every local form persists forever. It means that realized actuality is not ontologically reset at each step.

5.4 Admissibility

Only continuations coherent with both the Constitutive Law and realized history may be candidates for actualization. The Selective Law therefore acts not on arbitrary possibilities, but on a local continuation cone carved out by coherence.

Definition 7 (Admissibility)**.** *A continuation of history is admissible when it is coherent with both the Constitutive Law and the realized past it would extend.*

Admissibility is therefore the point at which the constitutive and selective roles touch. Constitutive constraint defines the admissible field; selective actualization acts only within it.

5.5 Local valuation

Suppose now that multiple admissible continuations are available. If the Selective Law contains no ordering principle, then actuality is brute. But the relevant ordering cannot be framed as a score assigned by inspecting remote unactualized futures.

A dynamic world must avoid two opposite failures. If it always favors bare preservation, it stagnates. If it always favors novelty as such, it explodes into unstable excess. Hence valuation must be dual. It must regard both what a continuation immediately opens and what it requires in order to become real without destabilizing what already exists.

Definition 8 (Intrinsic generative affordance)**.** *The intrinsic generative affordance of a continuation is the degree to which its present structure immediately opens further coherent continuation space.*

Intrinsic generative affordance is not a forecast produced by simulating remote futures. It is the locally readable generative room already encoded in the continuation itself and in the immediate continuation field it opens.

Definition 9 (Stabilization burden)**.** *The stabilization burden of a continuation is the additional coordination, integration, and maintenance required to make it actual without destabilizing realized history.*

Intrinsic generative affordance without stabilization collapses into uncontrolled branching. Stabilization without intrinsic generative affordance collapses into sterile conservation. A genuine selective rule must therefore weigh both.

5.6 Critical sufficiency

Even dual valuation is not enough. A world that always chose the abstractly largest continuation would collapse into gratuitous composite leaps. Distinct admissible gains could be bundled into one oversized transition, and actuality would lose articulation.

Therefore the Selective Law cannot be a rule of simple maximization. It must be *critical*: actuality should prefer what is sufficient *here and now*, not what is merely largest in the abstract.

Definition 10 (Critical sufficiency)**.** *A continuation is sufficient in context when its intrinsic*

generative affordance is strong enough, relative to its stabilization burden, to justify actualization at the present stage of history.

Critical sufficiency introduces a context-sensitive threshold. Not every positive continuation actualizes. Only those strong enough to justify integration under present conditions can do so.

5.7 Constructive primeness

Among sufficient continuations, actuality must prefer those that are not merely artificial bundles of several independently sufficient moves. Otherwise criticality still leaves room for arbitrary packing, and history again loses articulation.

Definition 11 (Constructive primeness)**.** *A continuation is constructively prime at a stage when it is not merely a gratuitous bundle of smaller continuations each of which would already be independently sufficient for actualization in that context.*

Constructive primeness is what prevents arbitrary packing. It enforces the granularity of becoming.

5.8 Coherent integration

Actualization must not end with choice alone. What is actualized must enter history in a way that changes the next field of admissible continuations. Without that, there is no cumulative reality, only repeated isolated choices.

Definition 12 (Coherent integration)**.** *A selective order exhibits coherent integration when the actualized continuation is irreversibly incorporated into realized history and thereby reshapes the future admissible field.*

Coherent integration turns local actualization into world-history rather than episodic choice. Deeper future structure then arises by recursive iteration of the same local law, not by precomputation of a fully surveyed future.

5.9 Canonical statement of the Selective Law

We may now state the selective role in canonical form.

Theorem 2 (Selective Law)**.** *Among the coherent continuations of realized history, actuality selects the least constructively prime continuation whose intrinsic generative affordance is sufficient to justify its stabilization burden in the present context, and irreversibly integrates it into the basis of further actualization.*

The statement is deliberately local. It does not require reality to inspect indefinitely remote futures before actualization occurs. Future structure arises through recursively updated local continuation cones.

Remark 1. *A compressed formulation is useful. One may summarize the selective role by saying:*

reality becomes by least sufficient locally generative coherent expansion. *The fuller theorem should be preferred whenever precision matters.*

5.10 Why these requirements are needed

The previous subsections stated the selective role positively. The present subsection strengthens the case by showing that each component is forced by the failure of nearby alternatives.

5.10.1 Without localizability: teleology or opacity

If the law ordered continuations by properties not encoded in the present continuation cone, it would have to consult unactualized remote futures before actuality could proceed. The result would be either teleology in formulation or hidden computation in implementation. Therefore localizability is necessary.

5.10.2 Without commitment: no evolving world

A world without commitment is not an evolving world but a sequence of disconnected replacements. Nothing that has happened genuinely belongs to what comes next. Identity through change disappears, and with it any meaningful notion of history. Therefore commitment is necessary.

5.10.3 Without admissibility: self-violation

A selective law without admissibility can actualize what the constitutive law has already ruled out. In that case the ontology contradicts itself: possibility is constrained at one level and unconstrained at another. Therefore admissibility is necessary.

5.10.4 Without valuation: brute actuality

If multiple coherent continuations are available and nothing orders them, then actuality is brute. The world becomes one possibility rather than another for no internal reason. Therefore valuation is necessary.

5.10.5 Without intrinsic generative affordance: stagnation

A world that values only preservation may remain stable, but it cannot be genuinely dynamic. It can only conserve what already exists. No principled route to novelty remains. Therefore valuation cannot reduce to stabilization burden alone.

5.10.6 Without stabilization burden: explosion

A world that values only immediately opened continuation space may constantly introduce difference, but at the cost of losing stability and identity. Novelty becomes self-undermining. Therefore valuation cannot reduce to intrinsic generative affordance alone.

5.10.7 Without critical sufficiency: oversized jumps

If actuality always preferred the largest available continuation rather than the least sufficient one, composite candidates could swallow transitions that were already sufficient on their own. History would advance by gratuitous macro-moves rather than articulated becoming. Therefore the law must be critical rather than naively maximal.

5.10.8 Without constructive primeness: gratuitous bundling

Suppose critical sufficiency is retained but constructive primeness is not. Then the world still has a threshold, but it may clear that threshold by arbitrary bundling of independently sufficient continuations. The threshold alone cannot protect ontological economy or articulated becoming. Therefore constructive primeness is necessary.

5.10.9 Without coherent integration: no world-history

Suppose a continuation is actualized but not coherently integrated. Then it does not alter the future admissible field. Actualization becomes episodic rather than cumulative, and no world-history forms. Therefore coherent integration is necessary.

5.10.10 Therefore

Any universe rich enough to sustain determinate, cumulative becoming must obey a Selective Law that is:

- cumulative,
- coherence-preserving,
- locally evaluable,
- valuational,
- critical,
- prime-selective,
- and integrative.

Taken together, these constraints force selective actualization into something close to the canonical statement given above. The exact later formalization may vary. The structural shape cannot vary far without the ontology collapsing into arbitrariness, stagnation, explosion, serial replacement, or computational opacity.

6 What the Framework Adds

Constitutive constraint and selective actualization are not rival explanations. They are complementary.

Constitutive constraint answers the question: *what can coherently be?* It defines the admissible order of form. In a broadly structural register, it concerns the realm of coherent candidates as such.

Selective actualization answers the question: *what becomes actual?* It defines the order of realized history through recursive local actualization. In a broadly process-oriented register, it concerns the transition from candidate structure to inhabited actuality.

A useful way to state the contribution is this: constitutive constraint gives the world its *order of form*, while selective actualization gives it its *order of becoming*. The novelty is not merely to contrast possibility with actuality in the abstract, but to argue that foundational explanation must assign these two questions to distinct explanatory roles. The first specifies the candidate space but does not order actualization within it. The second orders actualization within the candidate space but does not create that space. Neither is reducible to the other without explanatory loss.

7 Scope, Limits, and What Is Deferred

A foundations paper gains strength by distinguishing necessity from proposal and proposal from speculation.

7.1 Argued as metaphysical necessity

The following claims are argued here as necessary for any stable and interesting reality:

1. Reality requires both a constitutive law and a selective law.
2. The constitutive law must at least require coherent realizability, equivalence-invariance, canonical executability, and local satisfiability of compatibility burdens.
3. The selective law must at least require localizability, commitment, admissibility, valuation, critical sufficiency, constructive primeness, and coherent integration.

These are the core philosophical results of the paper.

7.2 Proposed as canonical formulation

The following stronger claims are proposed as the best current articulation of the foregoing necessities:

1. The Constitutive Law is best stated as the requirement that a universe be a coherent, equivalence-invariant, canonically executable structure whose identity and compatibility conditions are locally satisfiable.
2. The Selective Law is best stated as the rule that actuality selects the least constructively prime coherent continuation whose intrinsic generative affordance is sufficient to justify its stabilization burden in context, and integrates it irreversibly into further history.

These formulations are stronger than the bare necessities because they package the minimal requirements into single canonical statements.

7.3 Deferred to later work

The following questions are intentionally left open here:

1. The exact mathematical form of valuation and local actualization.
2. Any concrete evaluator, threshold equation, or dynamical mechanism that might instantiate the Selective Law.
3. The formal status of arithmetic and the natural numbers, beyond the weak assumption that cumulative actualization can be discussed stagewise.
4. Probabilistic or amplitude-like refinements of actualization.
5. Cosmological, physical, or phenomenological applications.
6. Any claim that a single human formalism uniquely and exhaustively captures the constitutive order.

This paper is therefore not a complete metaphysical system. It is the identification of the two irreducible tasks any such system must perform, together with the strongest presentation-independent articulation currently available.

8 Conclusion

The paper has argued for a simple but demanding thesis. A universe cannot be constituted by form alone, because form alone never explains actuality. Nor can it be constituted by actuality alone, because actuality without admissible form is unintelligible. A determinate world requires both a constitutive law and a selective law.

The constitutive order determines what can coherently count as possible. The selective order determines how coherent possibilities are locally actualized and integrated into history. At the foundational level there is no third irreducible role. Every further primitive law will either constrain coherent possibility or govern actualization from within it.

The resulting picture may be summarized in one sentence:

> Reality is neither mere form nor brute fact. It is coherent possibility under a law of local actualization.

Or, still more compactly:

> A universe exists only where coherent form and local actualization meet.

That is the paper's final claim. If it is right, then the deepest problem for metaphysics is not to choose between mathematics and world, nor between possibility and actuality, but to understand why reality requires both – and why the two laws that govern them must have the form they do.

References

[1] Max Tegmark, *The Mathematical Universe*, Foundations of Physics **38** (2008), 101–150.

[2] Per Martin-Löf, *Intuitionistic Type Theory*, Bibliopolis, Naples, 1984.

[3] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, Princeton, 2013.